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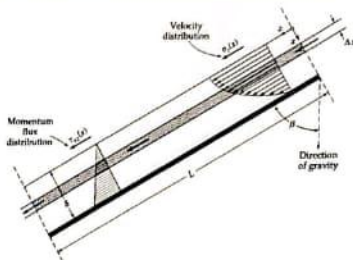
TRANSPORT PHENOMENA I, CHE 311

QUIZ 1

CLOSED BOOK, TIME: 15 MINUTES

The velocity profile of a liquid film flowing down an inclined flat plate is given by:

$$v_z = \frac{\rho g \cos \beta \delta^2}{2\mu} \left(1 - \frac{x^2}{\delta^2}\right)$$



- 1- Write down an equation for the z-component of the mass flux as a function of x.
- 2- Calculate the viscous tension on the gas-liquid and the solid-liquid interfaces.

All your answers must be in terms of the following constant parameters.

μ and ρ : viscosity and density of the liquid

δ : film thickness

β : angle of the flat plate relative to the direction of gravity

g : gravitational constant

Newton's law of viscosity: $|\tau| = -\mu(|\nabla v| + |\nabla v|^T)$

BSL textbook notation: $\tau = -\mu(\nabla v + (\nabla v)^T)$

$$v_z = \frac{\rho g \cos \beta \delta^2}{2\mu} \left(1 - \frac{x^2}{\delta^2}\right)$$

1) $J_z(x) = \rho v_z(x)$

$$J_z(x) = \frac{\rho^2 g \cos \beta \delta^2}{2\mu} \left(1 - \frac{x^2}{\delta^2}\right) \quad \checkmark \quad \underline{50}$$

2) $T_{xz} = -\mu \frac{dv_z}{dx}$

$$\frac{dv_z}{dx} = \frac{\rho g \cos \beta \delta^2}{2\mu} \left(-\frac{2x}{\delta^2}\right) = -\frac{\rho g \cos \beta}{\mu} x$$

$$T_{xz}(x) = -\mu \left(-\frac{\rho g \cos \beta}{\mu} x\right) = \rho g \cos \beta x \quad \checkmark$$

$$T_{xz}(\text{gas-liquid}) = 0$$

At $x = \delta$ $T_{xz}(\text{solid-liquid}) = \rho g \cos \beta \delta \quad \checkmark$

$$\text{Mass Flux} = J_z(x) = \frac{\rho^2 g \cos \beta \delta^2}{2\mu} \left(1 - \frac{x^2}{\delta^2}\right)$$